



UNIVERSITAT
POLITÈCNICA
DE VALÈNCIA

Aprendizaje Automático Probabilístico

Probabilidad: modelos multivariados (actividades)

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3.1. Distribuciones conjuntas para v.a. múltiples

3.1.1. Covarianza

► Halla la covarianza entre las variables aleatorias X e Y :

$p(X, Y)$	$Y = 1$	$Y = 2$	$Y = 3$	$p(X)$
$X = 1$	0.25	0.25	0	0.5
$X = 2$	0	0.25	0.25	0.5
$p(Y)$	0.25	0.5	0.25	1.0

$$\mathbb{E}[X] = 1 \cdot 0.5 + 2 \cdot 0.5 = 1.5$$

$$\mathbb{E}[Y] = 1 \cdot 0.25 + 2 \cdot 0.5 + 3 \cdot 0.25 = 2$$

$$\text{Cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

$$\begin{aligned} &= (1 - 1.5)(1 - 2)0.25 + (1 - 1.5)(2 - 2)0.25 + (1 - 1.5)(3 - 2)0 \\ &+ (2 - 1.5)(1 - 2)0 + (2 - 1.5)(2 - 2)0.25 + (2 - 1.5)(3 - 2)0.25 \\ &= 0.125 + 0 + 0 + 0 + 0 + 0.125 = 0.25 \end{aligned}$$

► Prueba que:

$$\mathbb{E}[\mathbf{x}\mathbf{x}^t] = \Sigma + \mu\mu^t$$

$$\begin{aligned}\Sigma &= \mathbb{E}[(\mathbf{x} - \mu)(\mathbf{x} - \mu)^t] \\&= \mathbb{E}[\mathbf{x}\mathbf{x}^t - \mathbf{x}\mu^t - \mu\mathbf{x}^t + \mu\mu^t] \\&= \mathbb{E}[\mathbf{x}\mathbf{x}^t] - \mathbb{E}[\mathbf{x}]\mu^t - \mu\mathbb{E}[\mathbf{x}]^t + \mu\mu^t \\&= \mathbb{E}[\mathbf{x}\mathbf{x}^t] - \mu\mu^t - \mu\mu^t + \mu\mu^t \\&= \mathbb{E}[\mathbf{x}\mathbf{x}^t] - \mu\mu^t\end{aligned}$$

3.1.2. Correlación

► Halla la correlación entre las variables aleatorias X e Y :

$p(X, Y)$	$Y = 1$	$Y = 2$	$Y = 3$	$p(X)$
$X = 1$	0.25	0.25	0	0.5
$X = 2$	0	0.25	0.25	0.5
$p(Y)$	0.25	0.5	0.25	1.0

$$\mathbb{E}[X] = 1.5$$

$$\mathbb{E}[Y] = 2$$

$$\mathbb{V}[X] = (1 - 1.5)^2 \cdot 0.5 + (2 - 1.5)^2 \cdot 0.5 = 0.25$$

$$\mathbb{V}[Y] = (1 - 2)^2 \cdot 0.25 + (2 - 2)^2 \cdot 0.5 + (3 - 2)^2 \cdot 0.25 = 0.5$$

$$\text{Cov}[X, Y] = 0.25$$

$$\text{corr}[X, Y] = \frac{\text{Cov}[X, Y]}{\sqrt{\mathbb{V}[X] \mathbb{V}[Y]}} = \frac{1/4}{\sqrt{1/8}} = \frac{\sqrt{2}}{2} = 0.71$$

3.1.3. Correlación nula no implica independencia

- Prueba que, si dos variables aleatorias X e Y son independientes, entonces $\text{Cov}[X, Y] = \text{corr}[X, Y] = 0$

Por independencia de X e Y (sec. 2.2.5.2),

$$\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$$

Por tanto,

$$\text{Cov}[X, Y] = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] = 0$$

$$\text{corr}[X, Y] = \frac{\text{Cov}[X, Y]}{\sqrt{\mathbb{V}[X] \mathbb{V}[Y]}} = 0$$

- Sean $X \sim U(-1, 1)$ e $Y = X^2$, obviamente dependiente de X .
Prueba que $\text{Cov}[X, Y] = \text{corr}[X, Y] = 0$. Pista: $\mathbb{E}[X^3] = 0$.

$$\begin{aligned}\text{Cov}[X, Y] &= \text{Cov}[X, X^2] \\ &= \mathbb{E}[X X^2] - \mathbb{E}[X] \mathbb{E}[X^2] \\ &= \mathbb{E}[X^3] - \mathbb{E}[X] \mathbb{E}[X^2] \\ &= 0 - 0 \cdot \mathbb{E}[X^2] \\ &= 0 = \text{corr}[X, Y]\end{aligned}$$

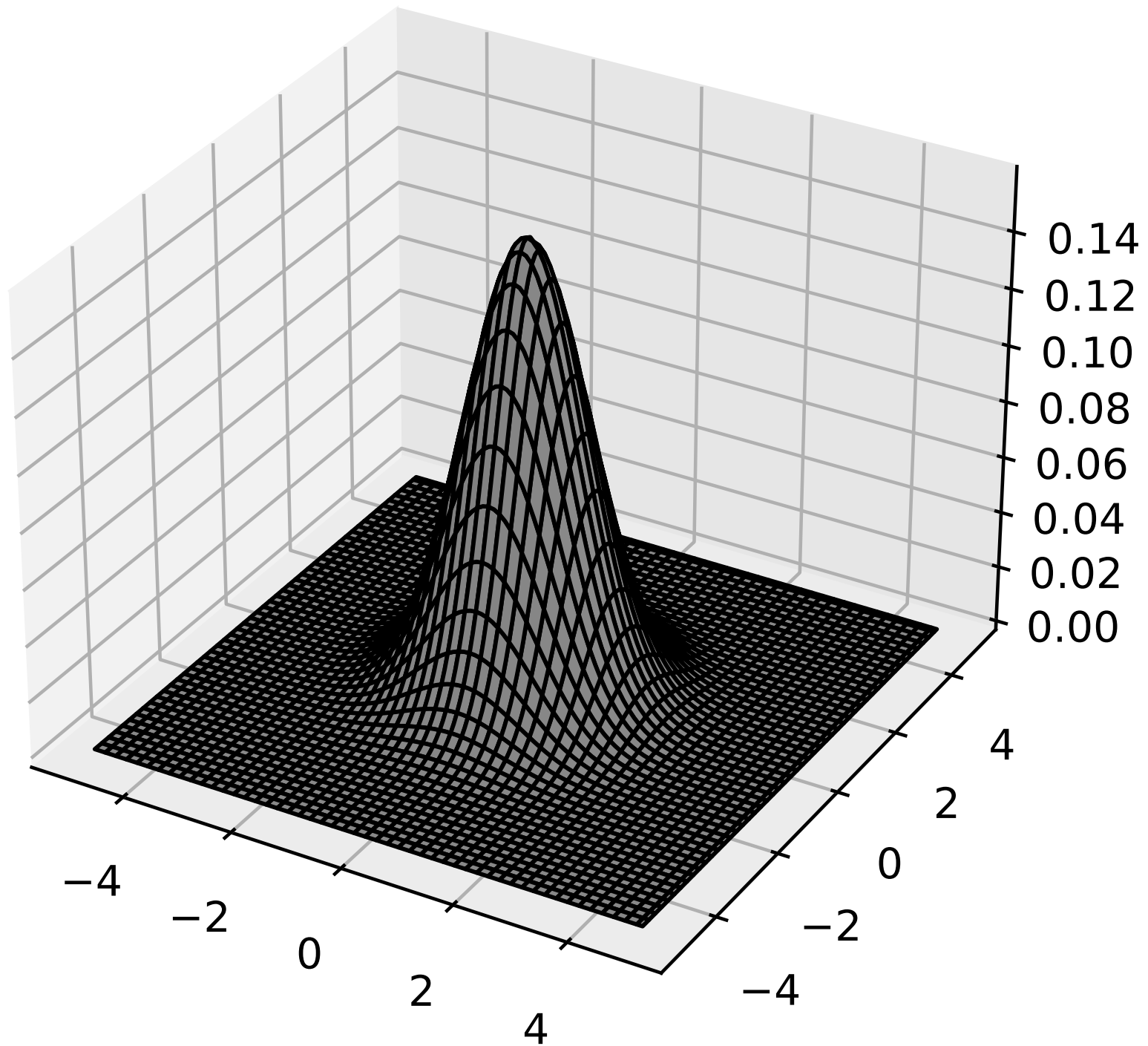
$$\mathbb{E}[X^3] = \frac{1}{2} \int_{-1}^1 x^3 dx = 0$$

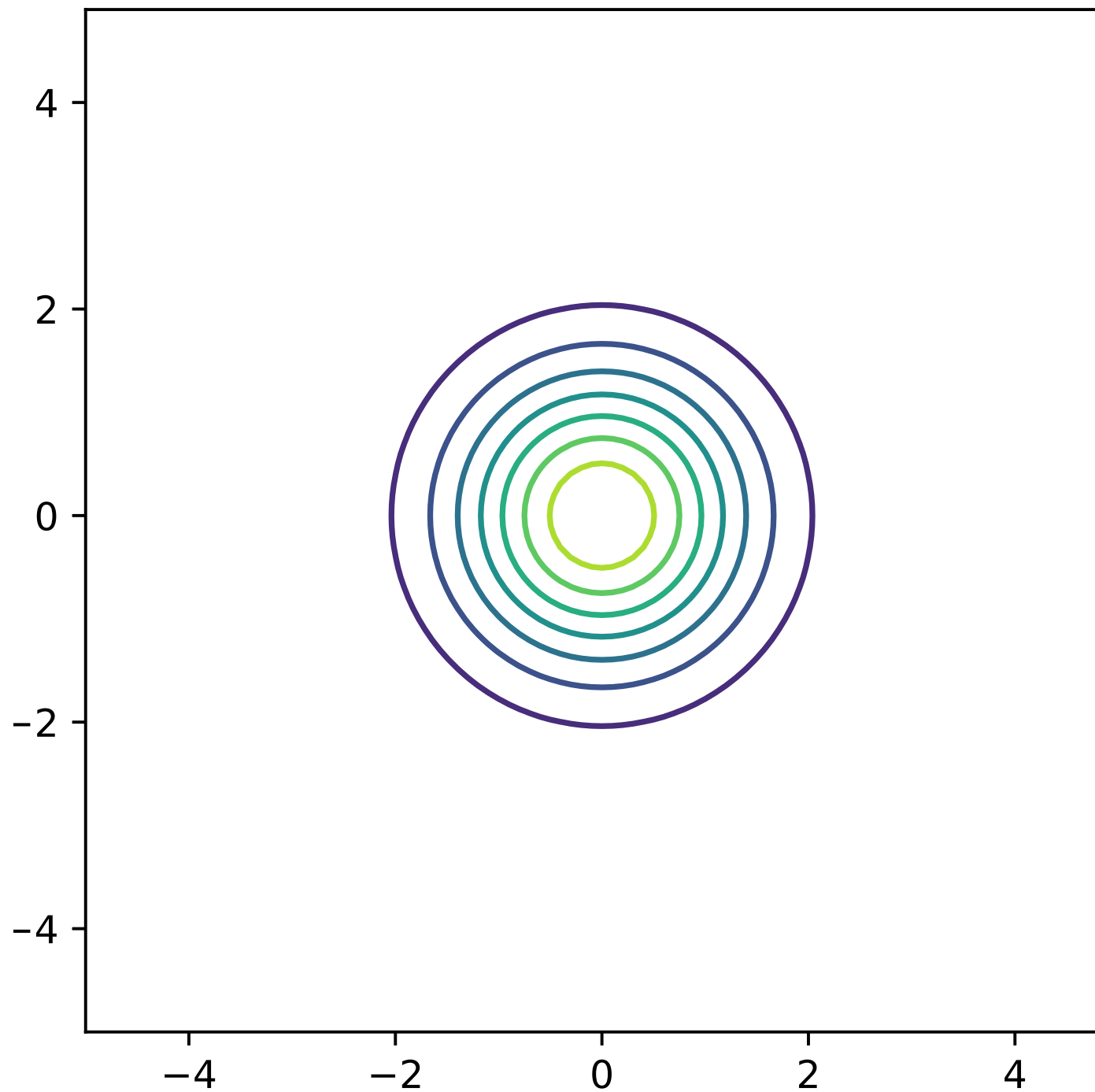
3.2. La Gaussiana (normal) multivariada

► Σ esférica: $\Sigma = \sigma^2 \mathbf{I}$, $\sigma^2 = 1$

3.2.N.py

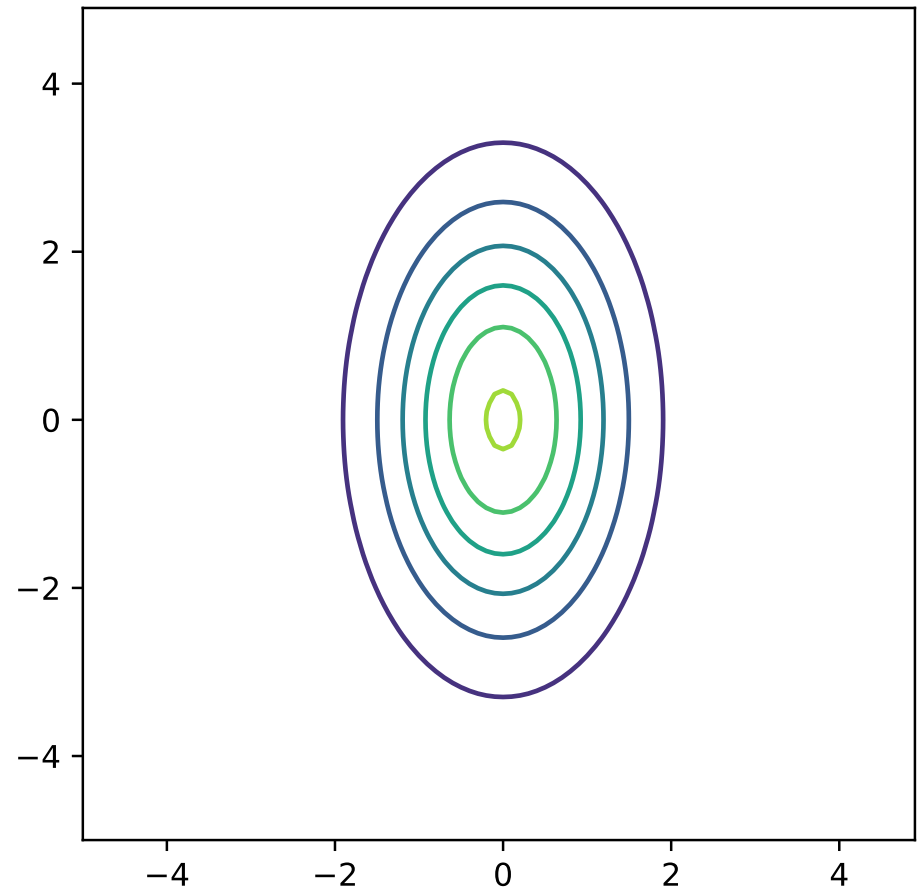
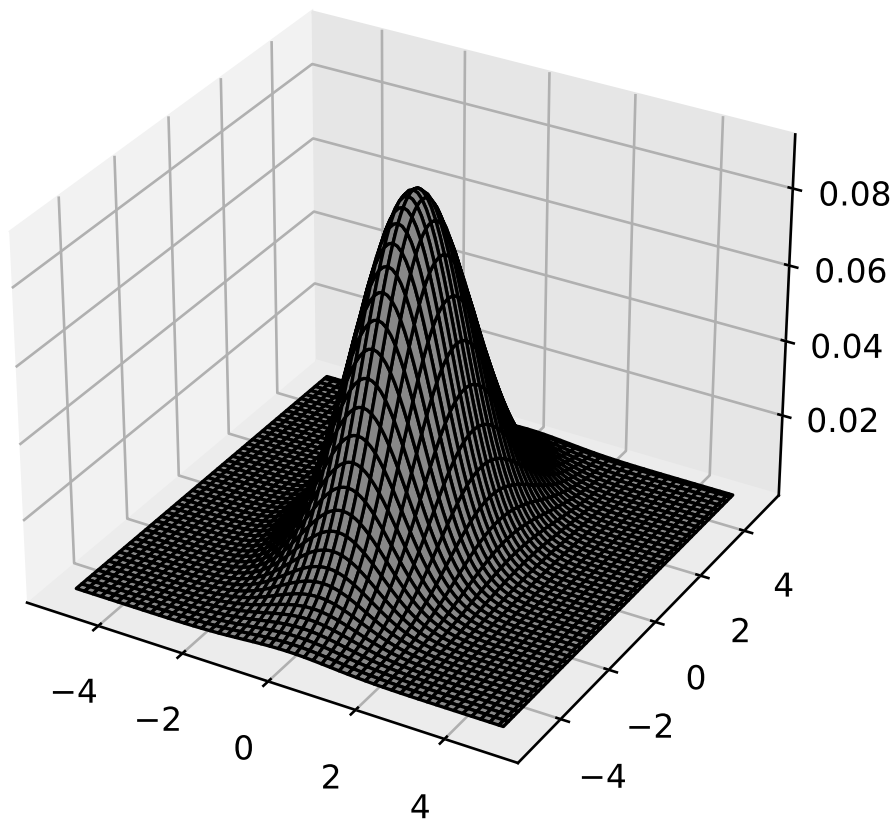
```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import multivariate_normal
4
5 m, S = [0, 0], [[1, 0], [0, 1]]
6 x, y = np.mgrid[-5:5:.1, -5:5:.1]
7 z = multivariate_normal(m, S).pdf(np.dstack((x, y)))
8
9 fig = plt.figure()
10 ax = fig.add_subplot(111, projection='3d')
11 ax.plot_surface(x, y, z, color='white',
12                ↪ edgecolor="black")
13
14 fig = plt.figure()
15 ax = fig.add_subplot(111, aspect='equal')
16 ax.contour(x, y, z)
17 plt.show()
```



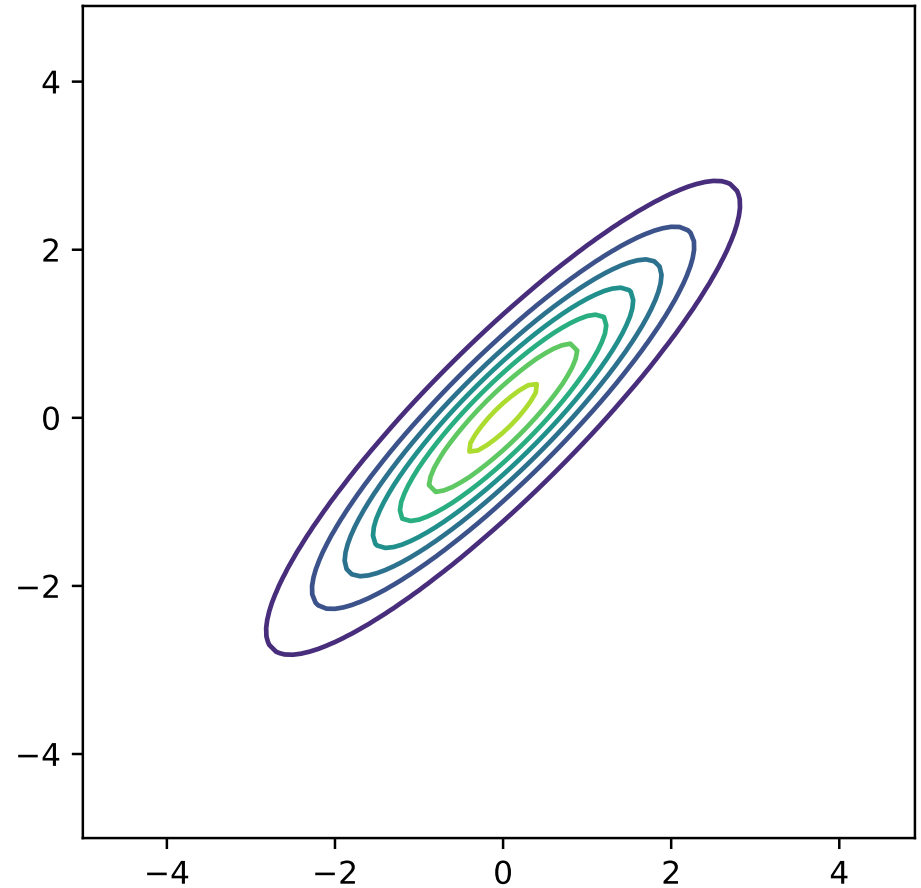
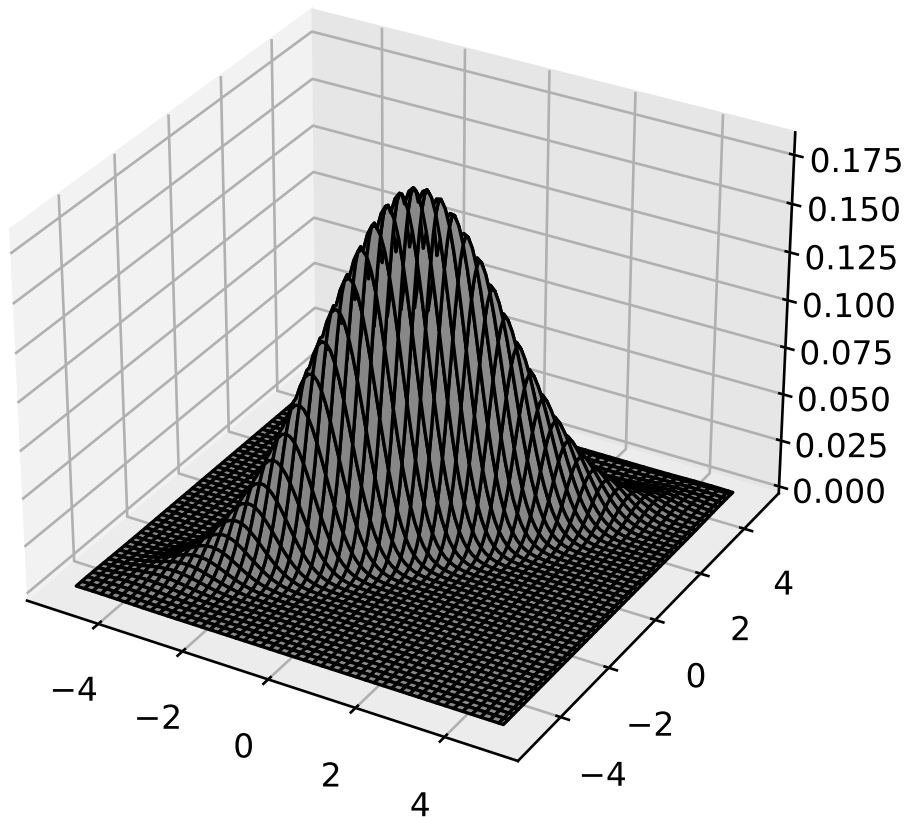
- Modifica el script anterior para el caso Σ *diagonal*:

$$\mu = 0 \quad y \quad \Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$$



- Modifica el script anterior para el caso Σ completa:

$$\mu = 0 \quad y \quad \Sigma = \begin{pmatrix} 2 & 1.8 \\ 1.8 & 2 \end{pmatrix}$$



- Descomposición en valores propios (espectral) de Σ : $\Sigma = U\Lambda U^t$

3.2.eigh.py

```

1 import numpy as np
2 S = np.array([[2, 1.8], [1.8, 2]])
3 La, U = np.linalg.eigh(S)
4 i = La.argsort()[::-1]; La = La[i]; U = U[:,i]
5 print(La, U, U @ np.diag(La) @ U.T)

```

3.2.eigh.out

```

1 [3.8 0.2] [[ 0.70710678 -0.70710678]
2 [ 0.70710678  0.70710678]] [[2.  1.8]
3 [1.8 2.  ]]

```

► Comprueba que
$$\begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 3.8 & 0 \\ 0 & 0.2 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} 2 & 1.8 \\ 1.8 & 2 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 3.8 & 0 \\ 0 & 0.2 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} 3.8\frac{\sqrt{2}}{2} & -0.2\frac{\sqrt{2}}{2} \\ 3.8\frac{\sqrt{2}}{2} & 0.2\frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 3.8\frac{1}{2} + 0.2\frac{1}{2} & 3.8\frac{1}{2} - 0.2\frac{1}{2} \\ 3.8\frac{1}{2} - 0.2\frac{1}{2} & 3.8\frac{1}{2} + 0.2\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 2 & 1.8 \\ 1.8 & 2 \end{bmatrix}$$

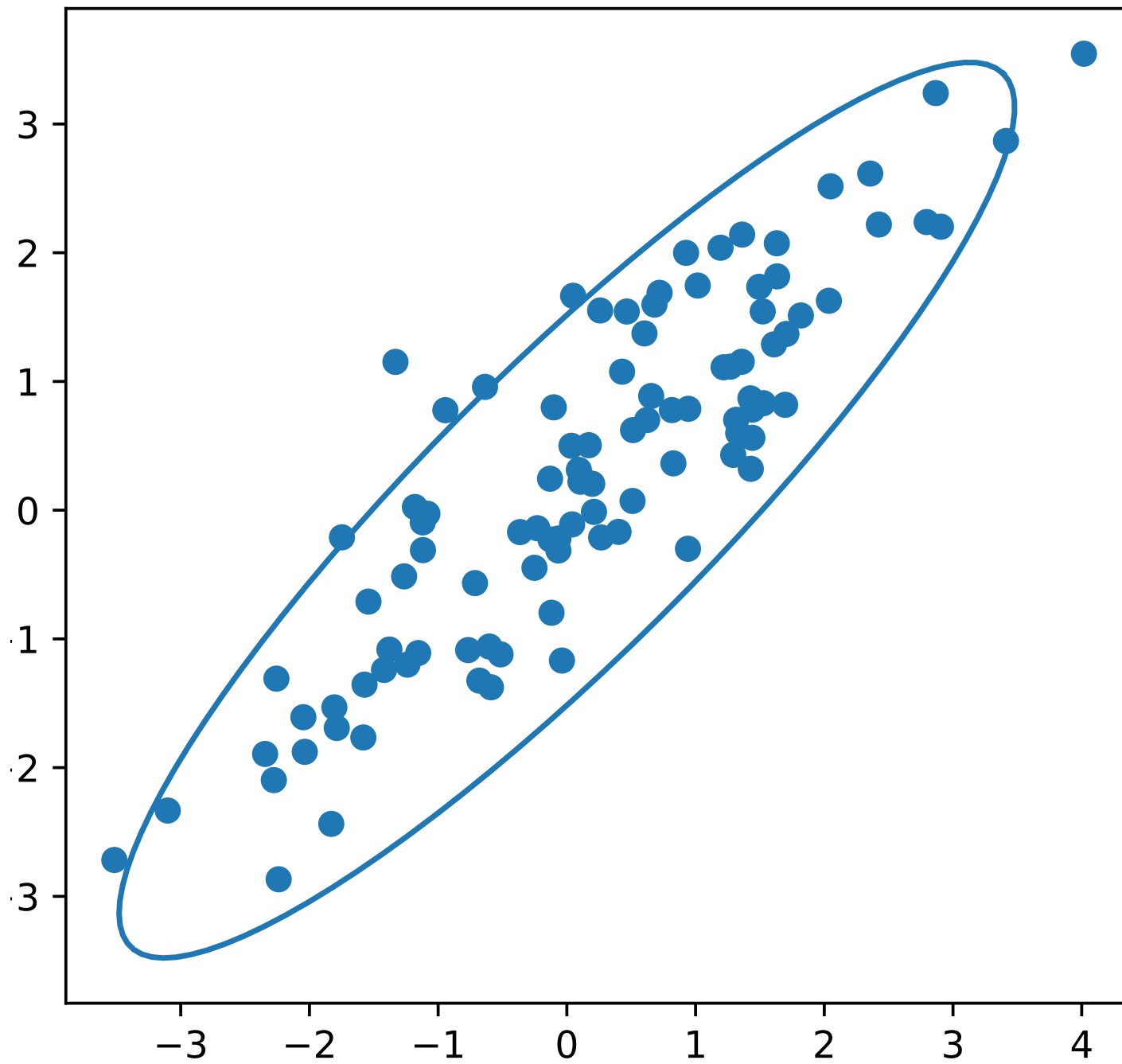
- Bola de Mahalanobis centrada en μ , de $\Sigma = \mathbf{U}\Lambda\mathbf{U}^t$ y radio r :

$$\begin{aligned}
& \{\mathbf{y} : \mathcal{B}_{\Sigma^{-1}}(\mathbf{y}, \mu) = r\} \quad (\text{frontera de la bola}) \\
&= \{\mathbf{y} : (\mathbf{y} - \mu)^t \Sigma^{-1} (\mathbf{y} - \mu) = r\} \\
&= \{\mathbf{y} : \mathbf{z} = \mathbf{y} - \mu, \mathbf{z}^t \Sigma^{-1} \mathbf{z} = r\} \\
&= \{\mathbf{y} : \mathbf{z} = \mathbf{y} - \mu, \mathbf{z}^t \mathbf{U} \Lambda^{-1} \mathbf{U}^t \mathbf{z} = r\} \\
&= \{\mathbf{y} : \mathbf{z} = \mathbf{y} - \mu, \underbrace{\mathbf{z}^t \mathbf{U} \Lambda^{-1/2}}_{\mathbf{x}^t} \underbrace{\Lambda^{-1/2} \mathbf{U}^t}_{\mathbf{x}} \mathbf{z} = r\} \\
&= \{\mathbf{y} : \mathbf{x} = \Lambda^{-1/2} \mathbf{U}^t (\mathbf{y} - \mu), \mathbf{x}^t \mathbf{x} = r\} \\
&\stackrel{2D}{=} \left\{ \mathbf{y} : \Lambda^{-1/2} \mathbf{U}^t (\mathbf{y} - \mu) = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix}, 0 \leq \theta \leq 2\pi \right\} \\
&= \left\{ \mathbf{y} : \mathbf{y} = \mu + \mathbf{U} \Lambda^{1/2} \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix}, 0 \leq \theta \leq 2\pi \right\}
\end{aligned}$$

▷ $\mu = \mathbf{0}$, $U = \begin{pmatrix} .71 & -.71 \\ .71 & .71 \end{pmatrix}$, $\Lambda = \text{diag}(3.8, .2)$ y $r = 2.45$ (95 %):

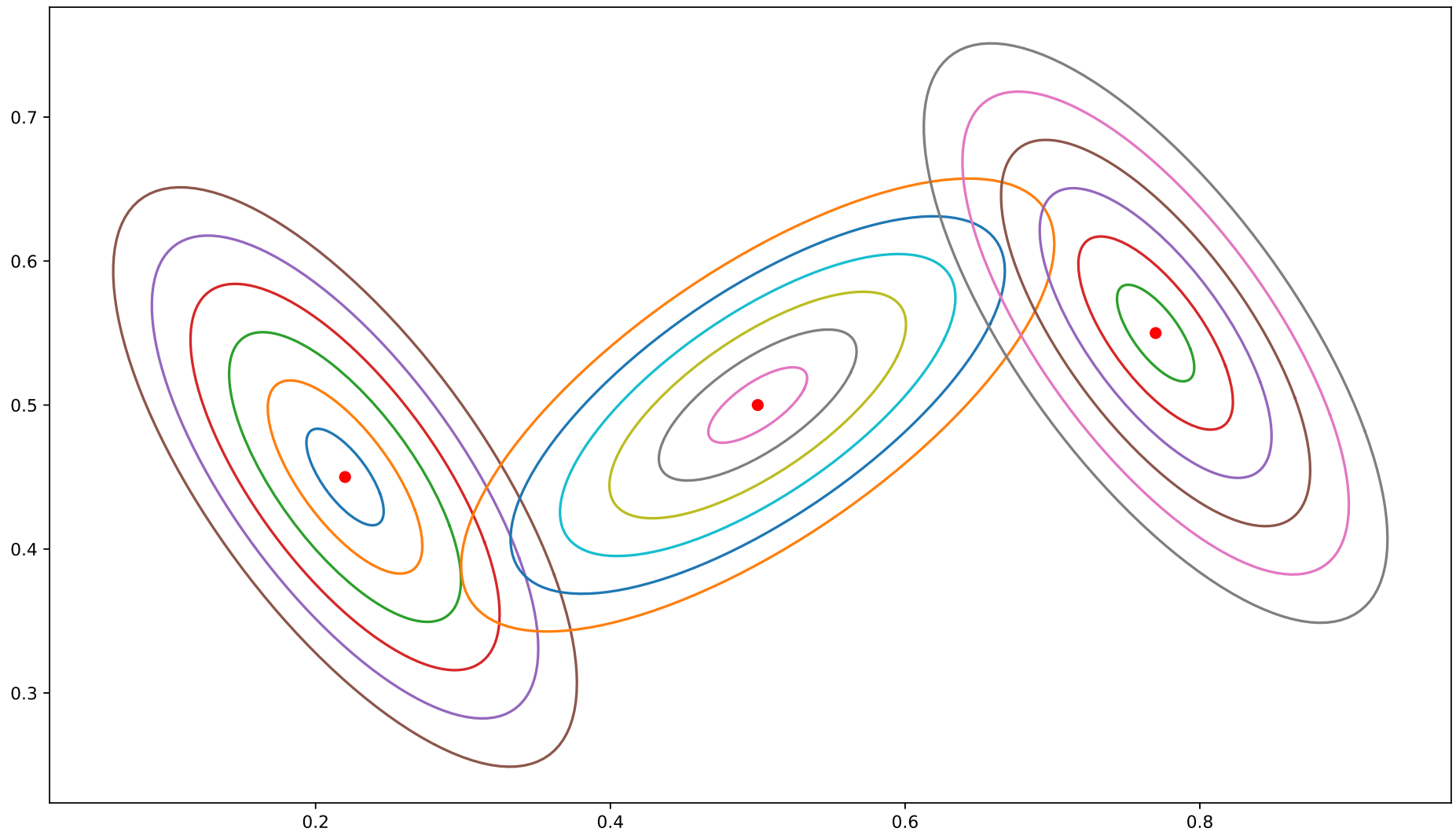
3.2.Mahalanobis.py

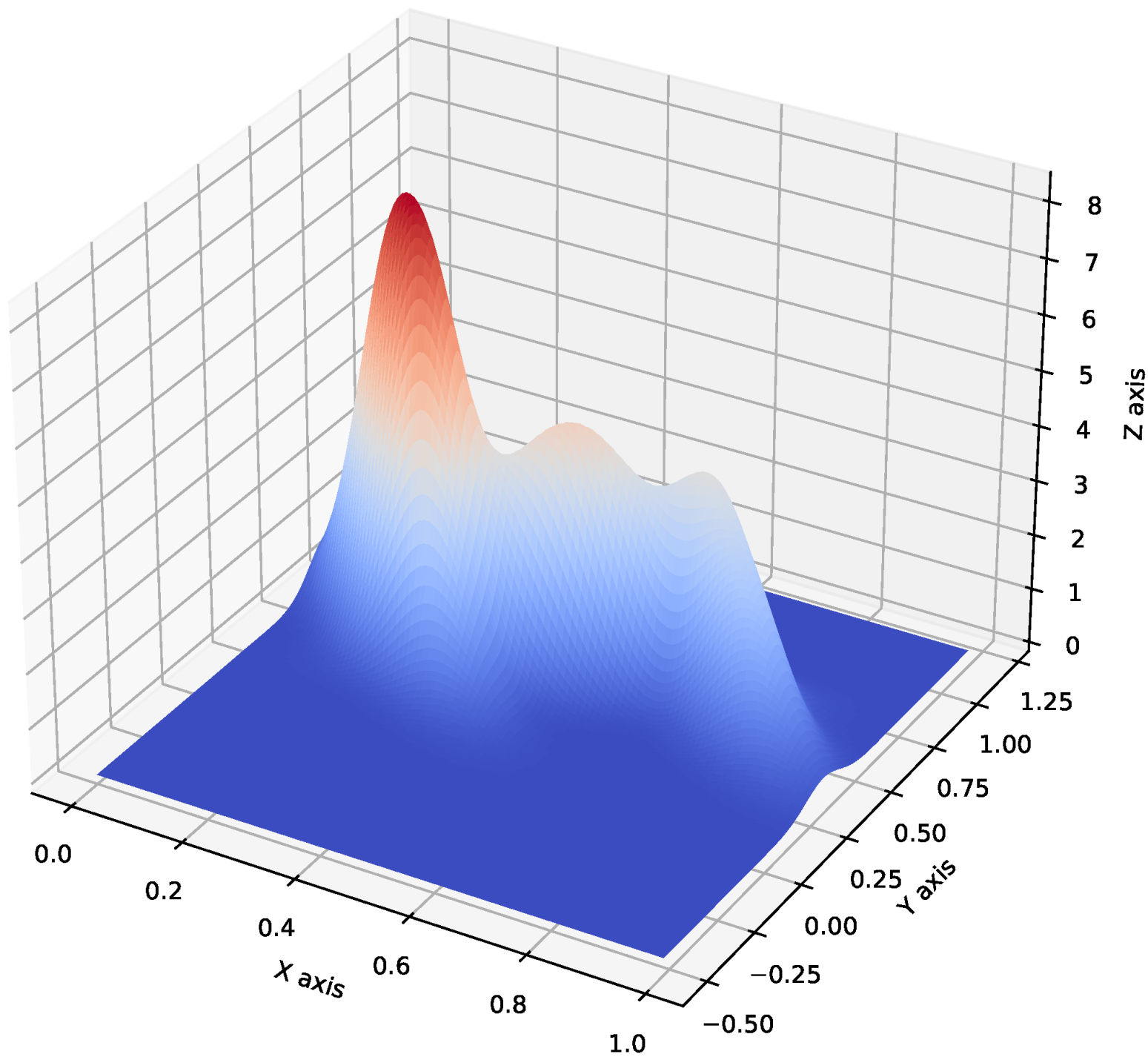
```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import multivariate_normal
4
5 U = np.array([[.71, -.71], [.71, .71]])
6 La = np.array([3.8, .2])
7 r = 2.45
8 t = np.linspace(0, 2*np.pi, 100)
9 x, y = U @ np.diag(np.sqrt(La)) @
    ↪ np.array([r*np.cos(t), r*np.sin(t)])
10 Y = np.vstack((x, y)).T
11 cov = U @ np.diag(La) @ U.T
12 X = multivariate_normal(cov=cov).rvs(100)
13 fig = plt.figure()
14 ax = fig.add_subplot(111, aspect='equal')
15 plt.plot(Y[:, 0], Y[:, 1])
16 plt.scatter(X[:, 0], X[:, 1])
17 plt.show()
```



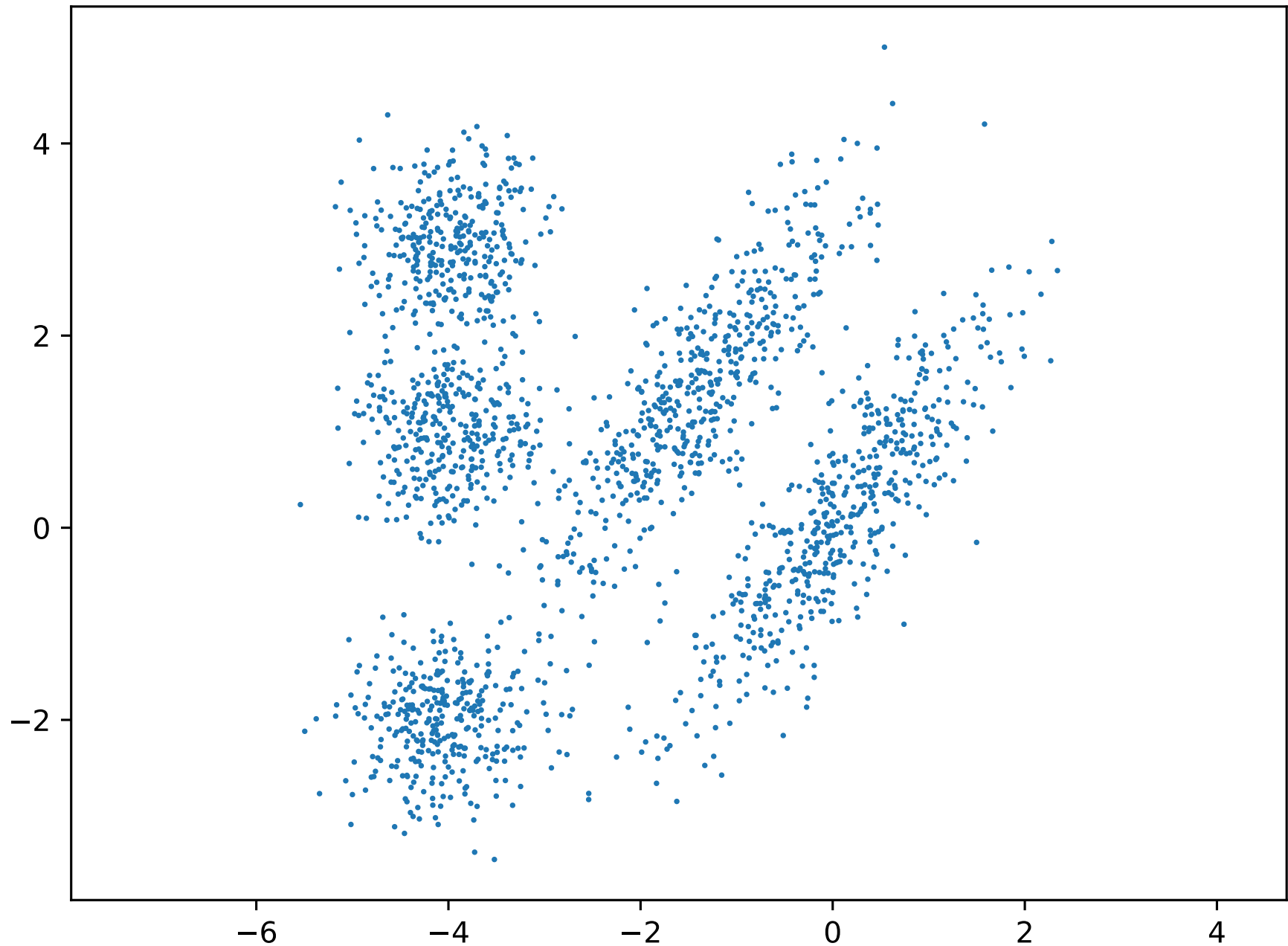
3.5. Modelos de mixtura

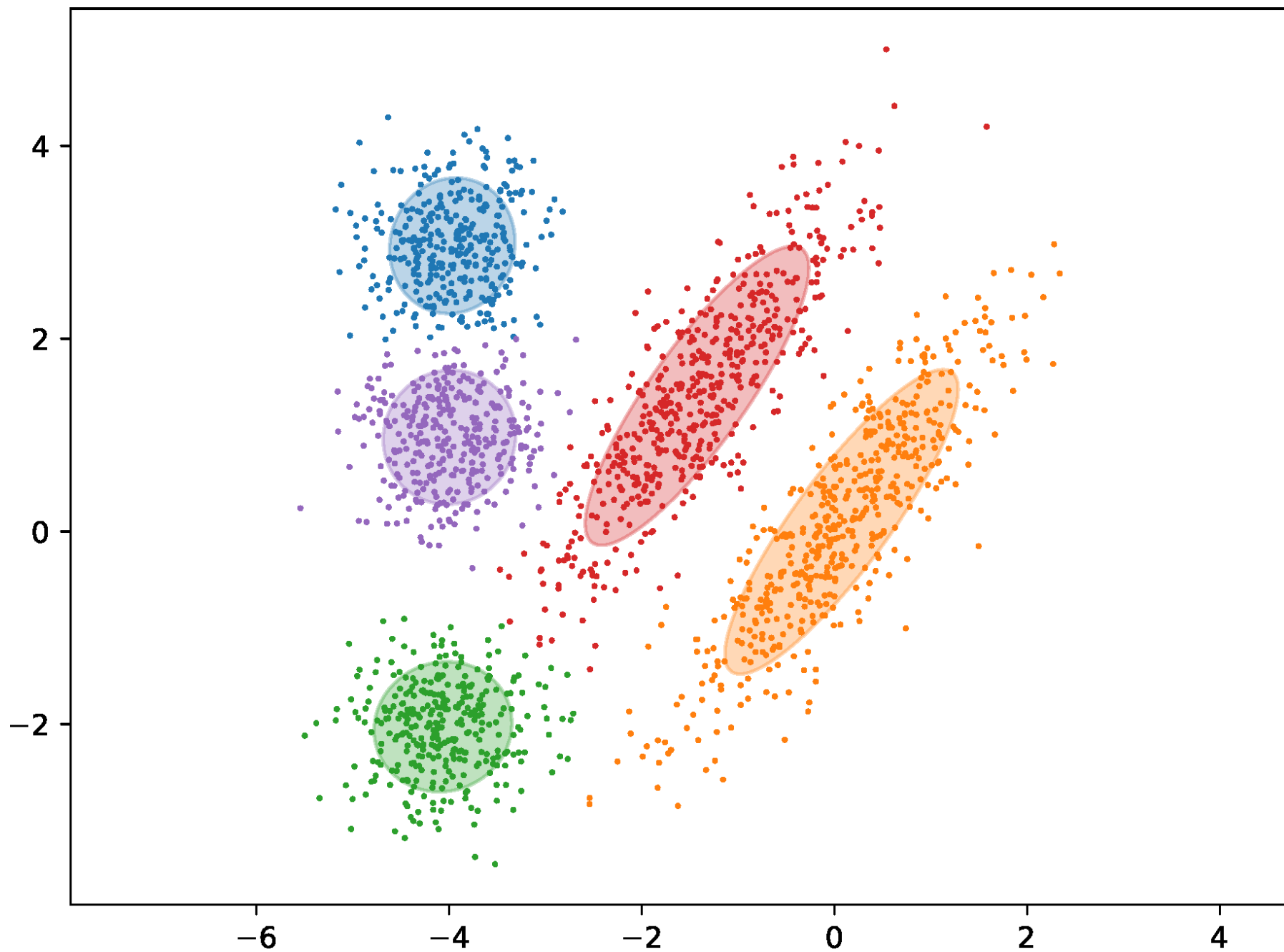
► Reproduce el cuaderno de la [Figura 3.11](#)





► Reproduce el cuaderno de la [Figura 3.12](#)





► Reproduce el cuaderno de la [Figura 3.13](#)

